

Smoothed Lv Distribution Based Three-Dimensional Imaging for Spinning Space Debris

Zhenyu Zhuo, Lan Du[✉], Senior Member, IEEE, Xiaofei Lu, Jian Chen[✉], and Zhuowei Cao

Abstract—Three-dimensional (3-D) imaging plays a vital role in the recognition of spinning space debris. However, the image may be blurred due to the range migration caused by fast rotation of space debris. Moreover, the image quality, which depends on the estimation accuracy of Doppler frequency and chirp rate of scattering centers, is influenced by cross-terms and sidelobes. In this article, we propose a novel 3-D imaging method based on smoothed Lv distribution (SLVD). Firstly, the selection criterion for best imaging time based on the time–frequency moment is proposed to guarantee that the echo is approximated as a linear frequency modulation (LFM) signal. Then, we operate the Khatri–Rao product on the centroid frequency and chirp rate (CFCR) representation and the range–Doppler (RD) image to obtain the 3-D image. To decrease the influence of range migration, we process a short time window during the RD imaging procedure. For cross-term and sidelobe suppression, the SLVD is proposed to obtain the CFCR representation by expressing the Lv distribution (LVD) in a convolution form and introducing a centroid frequency window. Experimental results verify the effectiveness of the proposed imaging method and the good performance of the proposed SLVD for cross-term and sidelobe suppression.

Index Terms—3-D imaging, cross-term and sidelobe suppression, range migration, smoothed Lv distribution (SLVD), space debris.

I. INTRODUCTION

SPACE debris has been a potential threat to satellites and spacecraft for their high-speed motion, and the detection and recognition of space debris have become critical for space security [1]. Inverse synthetic aperture radar (ISAR), with its all-day, all-weather operation ability, and long-range detection capability, can estimate the motion and structural characteristics of space targets and has been widely used for space target recognition. In a conventional 2-D ISAR image, the target is projected onto the imaging plane [2], [3]. However, the space debris is usually non-cooperating and has spinning motion. The scattering centers on the spinning space debris are rotating around the spinning axis. It is difficult to

obtain their real size in the cross-range, and the 2-D imaging method cannot obtain the height of scattering centers.

Compared with the 2-D ISAR image, the 3-D image can obtain more motion and structure features. The existing 3-D imaging methods are mainly based on the interferometric ISAR (InISAR) technique [4], [5]. The 3-D Cartesian coordinates of scattering centers and the 3-D motion parameters can be estimated by using multistatic radars to observe the target. The InISAR technique suffers from high hardware complexity for applying multiple sensors. Apart from the InISAR technique, there are other monostatic imaging methods. In [6]–[9], the 3-D coordinates of dominant scattering centers are estimated from the range–slow time image by a parametric algorithm, such as the generalized Radon transform [6], Hough transform [7], [8], and the modified particle swarm optimization [9]. These methods have good noise robustness and tolerate the missing data, but they have to establish accurate parametric models and suffer a huge computational burden for searching in the large parameter space. In [10] and [11], the 3-D reconstruction of dominant scattering centers is obtained via the factorization-based algorithm from the associated trajectories. It is unnecessary to estimate the motion parameters, but the accurate association of scattering center trajectories is the key challenge. In [12] and [13], slant range, Doppler frequency, and chirp rate form a 3-D space, and these parameters can be estimated by the linear frequency modulation (LFM) parameter estimation method. The visual 3-D image can be obtained by estimating these parameters. In [12], shipborne echoes in the selected imaging time are proven to be LFM signals. The REACT algorithm, which combines the RELAX algorithm with adaptive chirplet transform (ACT), is proposed to estimate the parameters of the chirp signal. In [13], fractional Fourier transform (FrFT) is applied to estimate the parameters of ladar echoes. After scaling, the 3-D structure of the space debris would be realized. However, since the pulse repetition frequency and the sampling rate of radar echoes are much lower than those of ladar echoes, a longer imaging time is required to ensure the chirp rate resolution in radar imaging. Due to the fast rotation of space debris, the range of scattering center is modulated by the fast rotation in the imaging time, which induces the range migration and makes the image blurred. Therefore, range migration is the key challenge for 3-D imaging of the spinning space debris. In the meanwhile, the LFM parameter estimation method is important to improve imaging quality.

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Many studies on LFM signals have focused on parameters estimation and can be divided into two categories, i.e., time–frequency representation (TFR) and frequency–chirp rate representation (FCR). In the TFR category, the short-time Fourier transform (STFT) displays the frequency in the time–frequency spectrograms. The STFT is free from cross-term interference but suffers from poor resolution [14]. The Wigner-Ville distribution (WVD) can achieve the signal concentration but suffers from the cross-terms [15]. Thus, many WVD-based methods, such as Radon–Wigner distribution [16] and Wigner–Hough transform [17], combine TFR with line detection to decline the interference of the cross-terms, but the line searching operation brings a huge computational burden. In the FCR category, the FrFT obtains the chirp rate by searching the rotation angle [18]. However, the rotation angle is not a physical attribute of an LFM signal, and interference terms around peaks are powerful due to the order search. The Lv distribution (LVD) based on the scaling principle provides improved estimation accuracy of centroid frequency and chirp rate (CFCR) [19]. The inevitable cross-terms and sidelobes deteriorate the performance of the LVD.

In this article, a new 3-D imaging method based on the smoothed LVD (SLVD) is proposed to obtain 3-D images of fast-spinning space debris. First, we propose a selection criterion for best imaging time based on the time–frequency moment. As the slant range of the scattering center is modulated sinusoidal, the smallest gradient of slant range and chirp rate occurs when the Doppler frequency is close to zero. To guarantee that the target's echo is approximated as a multicomponent LFM signal, we choose the best imaging time when the time–frequency moment is the smallest. Then, slant range, Doppler frequency, and chirp rate form the 3-D imaging space. In this article, we operate the Khatri–Rao product on the CFCR representation and the range-Doppler (RD) image to obtain the 3-D image. We add a short time window process during the RD imaging procedure to decrease the influence of range migration. In addition, the SLVD is proposed to obtain the CFCR representation. The proposed SLVD expresses the LVD in a convolution form and masks a centroid frequency window for cross-term and sidelobe suppression. Therefore, the SLVD can improve estimation accuracy compared with the LVD. After scaling, the 3-D Cartesian coordinates of scattering centers together with the size of the target are estimated.

The main contributions of this article are summarized as follows.

- 1) We propose a novel 3-D imaging method for fast-spinning space debris, in which the Khatri–Rao product is operated on the CFCR representation and the RD image to obtain the 3-D image. Electromagnetic experimental results demonstrate that this method is capable of obtaining 3-D image for fast-rotating space debris and acquiring a more accurate estimation of target size.
- 2) We propose a selection criterion for imaging time and use a short time window to decrease the influence of range migration. In the selected imaging time, the slant range is approximately invariable. The short time

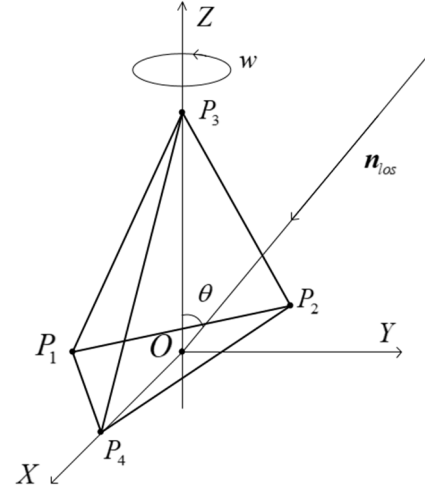


Fig. 1. Geometric model of space debris.

window process decreases the influence of range migration by shortening imaging time during the RD imaging procedure.

- 3) We propose the SLVD, which can effectively suppress the cross-terms and sidelobes. We discuss the principle of suppression with the analytical formula. And a numerical experiment demonstrates the effectiveness of cross-term and sidelobe suppression.

The organization of this article is as follows. The geometric model and echo model of space debris are given in Section II. The proposed imaging method based on SLVD is presented in Section III. The analysis of SLVD and detailed experimental results are shown in Section IV. Section V concludes this article.

II. GEOMETRIC MODEL AND SIGNAL MODEL

In this section, the geometric model and signal model of space debris are introduced. The scattering centers on the spinning space debris are rotating around the spinning axis. Fig. 1 shows the geometric model in the coordinate system $O - XYZ$, where the z -axis is the spinning axis, and the radar line-of-sight (rLOS) is on the YOZ plane. θ denotes the angle between the rLOS and the z -axis, and the rLOS vector is expressed as $\mathbf{n}_{los} = (0, \sin \theta, \cos \theta)$. Suppose that the space debris consists of N_0 scattering centers, and the 3-D coordinate of the n th scattering center in the centroid imaging time is $P_n = [x_n, y_n, z_n]^T$.

According to the motion model and the signal model in [13] and [20], the instantaneous slant range of the scattering center is expressed as the inner product between the rLOS vector and the coordinate vector

$$r_n(t_m) = (y_n \cos \omega t_m + x_n \sin \omega t_m) \sin \theta - z_n \cos \theta \quad (1)$$

where ω is the rotational angle velocity, and t_m is the slow time.

According to [21], the high-frequency scattering responses of a target can be described as a sum of responses from

individual scattering centers approximately. The echoes in the range frequency-slow time domain are expressed as

$$s_r(\hat{t}, t_m) = \sum_{n=1}^N \exp(j4\pi(u\hat{t} + f_c) \times [r_n(0) + r'_n(0)t_m + r''_n(0)t_m^2]) \quad (2)$$

where r'_n and r''_n denote the first and second derivative of $r_n(t_m)$, respectively. Here, $t_m = 0$ is the centroid imaging time. Therefore, the echo of the single frequency can be modeled as a multicomponent linear frequency signal as

$$s_r(t_m) = \sum_{n=1}^N \exp(j2\pi[2r_n(0)/\lambda + f_{d,n}t_m + K_{d,n}t_m^2]) \quad (3)$$

where λ denotes the wavelength, and the centroid Doppler frequency of the n th scattering center is

$$f_{d,n} = \frac{2}{\lambda}(wx_n \sin \theta - w^2 y_n \sin \theta t_m). \quad (4)$$

The chirp rate of the n th scattering center can be obtained by

$$K_{d,n} = \frac{df_{d,n}}{dt_m} = -\frac{2}{\lambda}w^2 y_n \sin \theta. \quad (5)$$

Slant range, Doppler frequency, and chirp rate can form a 3-D space. Combining (1), (4), and (5), we summarize the relationship between these parameters and the 3-D Cartesian coordinates of scattering centers

$$[x_n, y_n, z_n]^T = \left[\frac{\lambda f_{d,n}}{2w \sin \theta}, -\frac{\lambda K_{d,n}}{2w^2 \sin \theta}, -\frac{1}{\cos \theta} \left(r_n(0) + \frac{\lambda K_{d,n}}{2w^2} \right) \right]^T. \quad (6)$$

After scaling 3-D image by (6), we can estimate 3-D Cartesian coordinates and the size of space debris. From (6), we can find that there are two unknown parameters that can be estimated by two observations in [13].

III. 3-D IMAGING METHOD BASED ON SLVD

According to Section II, we can obtain the 3-D image and coordinates by estimating the slant range, Doppler frequency, and chirp rate. However, the fast rotation of space debris yields the range migration which makes the image blurred. In the meanwhile, imaging quality is influenced by cross-terms and sidelobes. In this section, we introduce a novel imaging method based on the proposed SLVD to decrease the influence of range migration and suppress cross-terms and sidelobes.

A. Proposed SLVD

1) *Review on the LVD*: For the explanation of the SLVD, we briefly review the definition and details of the LVD. More information about the LVD is available in [19]. First, a continuous infinite-time multicomponent LFM signal is considered as

$$x(t) = \sum_{i=1}^K A_i \exp(j2\pi f_i t + j\pi \gamma_i t^2), \quad -T \leq t \leq T \quad (7)$$

where K is the number of signal components and T is the time duration. A_i , f_i , and γ_i denote the constant amplitude, centroid frequency, and chirp rate of the i th signal component, respectively. Its continuous-time parametric symmetric instantaneous autocorrelation function (PSIAF) is expressed as

$$R_x(t, \tau) = x\left(t + \frac{\tau + a}{2}\right) x^*\left(t - \frac{\tau + a}{2}\right) = \sum_{i=0}^{K-1} A_i^2 \exp(j2\pi f_i(\tau + a) + j2\pi \gamma_i(\tau + a)t) + \text{Cr} \quad (8)$$

where τ is the lag variable, $*$ denotes complex conjugation, a denotes the time-delay parameter related to the scaling operator, and Cr denotes cross-terms. A scaling operator Γ with respect to (t, τ) is as follows

$$\Gamma[R_x(t, \tau)] = R_x\left(\frac{t_n}{h(\tau + a)}, \tau\right) = \sum_{i=0}^{K-1} A_i^2 \exp\left(j2\pi f_i(\tau + a) + j2\pi \frac{\gamma_i}{h} t_n\right) + \Gamma(\text{Cr}) \quad (9)$$

where h is a scaling factor and $t_n = (\tau + a)ht$ is the scaled time. Then, the fast Fourier transform (FFT), which is denoted as $\mathcal{F}$ in the terms of τ , is operated on (9) to obtain the scaled time-centroid frequency (STCF) representation as

$$L'(t_n, f) = \mathcal{F}_\tau(\Gamma[R_x(t, \tau)]) = \sum_{i=0}^{K-1} A_i^2 \delta(f - f_i) \exp\left(j2\pi \frac{\gamma_i}{h} t_n\right) \exp(j2\pi f_i a) + \mathcal{F}_\tau(\Gamma[\text{Cr}]). \quad (10)$$

And the CFCR representation can be obtained by FFT operation in the terms of t_n as

$$L(\gamma, f) = \mathcal{F}_{t_n}(L'(t, f)) = \sum_{i=0}^{K-1} A_i^2 \delta(f - f_i) \delta\left(\gamma - \frac{\gamma_i}{h}\right) \exp(j2\pi f_i a) + \mathcal{F}_\tau(\mathcal{F}_{t_n}(\Gamma[\text{Cr}])). \quad (11)$$

It is obvious that centroid frequencies and chirp rates of all signal components can be obtained by peak locations from CFCR domain in (11). According to (11) and the analysis in [19], there are cross-terms in the CFCR domain. In addition, the sidelobes, which are inspired by the signal noise and the dominating signal components, are appeared in CFCR domain. To solve these problems, we propose the SLVD for cross-term and sidelobe suppression.

2) *Principle of the Proposed SLVD*: The proposed SLVD expresses the LVD into the convolution form. Then, a centroid frequency window is introduced to suppress the cross-terms

and sidelobes. Suppose two signals as

$$\begin{aligned} u(t, \tau) &= x\left(t + \frac{\tau'}{2}\right) \\ &= \sum_{i=0}^{K-1} A_i \exp\left(j2\pi f_i\left(t + \frac{\tau'}{2}\right) + j\pi \gamma_i\left(t + \frac{\tau'}{2}\right)^2\right) \end{aligned} \quad (12)$$

$$\begin{aligned} v(t, \tau) &= x^*\left(t - \frac{\tau'}{2}\right) \\ &= \sum_{i=0}^{K-1} A_i \exp\left(-j2\pi f_i\left(t - \frac{\tau'}{2}\right) - j\pi \gamma_i\left(t - \frac{\tau'}{2}\right)^2\right) \end{aligned} \quad (13)$$

where $\tau' = \tau + a$. Applying the scaling operator Γ to these signals, we have

$$u(t_n, \tau) = \Gamma[u(t, \tau)] = x\left(\frac{t'_n}{\tau'} + \frac{\tau'}{2}\right) \quad (14)$$

$$v(t_n, \tau) = \Gamma[v(t, \tau)] = x^*\left(\frac{t'_n}{\tau'} - \frac{\tau'}{2}\right) \quad (15)$$

where $t'_n = ht\tau'$. By performing FFT on (14) and (15) with respect to τ , we have

$$U(t_n, f) = \int_{-T}^T x\left(\frac{t'_n}{\tau'} + \frac{\tau'}{2}\right) \exp(-j2\pi f \tau) d\tau \quad (16)$$

$$V(t_n, f) = \int_{-T}^T x^*\left(\frac{t'_n}{\tau'} - \frac{\tau'}{2}\right) \exp(-j2\pi f \tau) d\tau. \quad (17)$$

Note that, because of the FFT operation on the mono-component signal, there is no cross-term in (16) and (17). Then, we can express (10) into the convolution form as

$$L'(t_n, F) = \frac{1}{2\pi} U(t_n, f) \otimes_f V(t_n, f) \quad (18)$$

where $\otimes_f$ denotes the convolution with respect to f .

The proof of (18) is proven in the following. We calculate the right side of (18). The convolution can be expressed as a finite interval of the form $-l \leq f \leq l$ and let $l \rightarrow \infty$

$$\begin{aligned} &U(t_n, f) \otimes_f V(t_n, f) \\ &= \frac{1}{2\pi} \lim_{l \rightarrow \infty} \int_{-l}^l U(t_n, f) V(t_n, F - f) df \\ &= \lim_{l \rightarrow \infty} \int_{-l}^l \left[\int_{-T}^T x\left(\frac{t'_n}{\tau'} + \frac{\tau'}{2}\right) \exp(-j2\pi f \tau) d\tau \right. \\ &\quad \cdot \left. \int_{-T}^T x^*\left(\frac{t'_n}{v'} - \frac{v'}{2}\right) \exp(-j2\pi (F - f)v) dv \right] df \\ &= \int_{-T}^T \int_{-T}^T x\left(\frac{t'_n}{\tau'} + \frac{\tau'}{2}\right) x^*\left(\frac{t'_n}{v'} - \frac{v'}{2}\right) \exp(-j2\pi Fv) \\ &\quad \cdot \left[\lim_{l \rightarrow \infty} \int_{-l}^l \exp(-j2\pi f(\tau - v)) df \right] d\tau dv \\ &= \int_{-T}^T \int_{-T}^T x\left(\frac{t'_n}{\tau'} + \frac{\tau'}{2}\right) x^*\left(\frac{t'_n}{v'} - \frac{v'}{2}\right) \\ &\quad \cdot \exp(-j2\pi Fv) \delta(\tau - v) d\tau dv \end{aligned} \quad (19)$$

where $\delta(\cdot)$ is the Dirac delta function and $v' = v + a$. Due to $-T \leq \tau \leq T$, $-T \leq v \leq T$, and the sifting property in [22],

we integrate (19) with respect to v

$$\begin{aligned} &\frac{1}{2\pi} U(t_n, f) \otimes_f V(t_n, f) \\ &= \int_{-T}^T x\left(\frac{t'_n}{\tau'} + \frac{\tau'}{2}\right) x^*\left(\frac{t'_n}{\tau'} - \frac{\tau'}{2}\right) \exp(-j2\pi F\tau) d\tau \\ &= \int_{-T}^T R_x(t_n, \tau) \exp(-j2\pi F\tau) d\tau = L'(t_n, F). \end{aligned} \quad (20)$$

Thus, (18) has been proven. Note that, when $t_n = 0$, (18) is similar to (2) in [23]. Inspired by this idea, we introduce the centroid frequency window $P_1(\theta)$ to the convention-form LVD in (18)

$$L'_s(t_n, F) = \frac{1}{2\pi} \int P_1(f) U\left(t_n, -\frac{F}{2} + f\right) V\left(t_n, \frac{F}{2} - f\right) df, \quad t_n = 0 \quad (21)$$

where

$$P_1(f) = \begin{cases} 1, & |f| < f_{th} \\ 0, & \text{others.} \end{cases} \quad (22)$$

f_{th} is the length of the centroid frequency window. By windowing the product in convolution (18) through the narrow window, cross-terms will be reduced or even completely removed. We analyze the cross-terms and their suppression with an example in Section IV. Then, according to (10), the peaks of the STCF are located at $f = f_i$. Therefore, when $t'_n \neq 0$, we can introduce the centroid frequency window $P_2(\theta)$

$$L'_s(t_n, F) = \frac{1}{2\pi} P_2(F) \int U\left(t_n, -\frac{F}{2} + f\right) V\left(t_n, \frac{F}{2} - f\right) df, \quad t_n \neq 0 \quad (23)$$

where $P_2(F) = L'_s(0, F)$. In summary, we can express the windowed convolution-form LVD as

$$\begin{aligned} &L'_s(t_n, F) \\ &= \begin{cases} \frac{1}{2\pi} \int P_1(f) U\left(t_n, -\frac{F}{2} + f\right) V\left(t_n, \frac{F}{2} - f\right) df, \\ \quad t_n = 0 \\ \frac{1}{2\pi} P_2(F) \int U\left(t_n, -\frac{F}{2} + f\right) V\left(t_n, \frac{F}{2} - f\right) df, \\ \quad t_n \neq 0. \end{cases} \end{aligned} \quad (24)$$

Operating the FFT on (24) yields the CFCR representation obtained by the proposed SLVD can be obtained

$$L_s(\gamma, F) = \mathcal{F}_{t_n}(L'_s(t_n, F)). \quad (25)$$

3) Implementation of SLVD: In this subsection, we discuss the discrete implementation for the SLVD. The key step of the implementation is the convolution operation. Firstly, the discrete form of the multicomponent signal can be expressed as

$$\begin{aligned} x(n) &= \sum_i x_i(n) \\ &= \sum_i A_i \exp(j2\pi f_i n T_s + j\pi \gamma_i (n T_s)^2), \quad -N \leq n \leq N \end{aligned} \quad (26)$$

where T_s denotes the sampling interval and the number of samples is $N = T/T_s$. To simplify the analysis, we ignore the cross-terms of the PSIAF in (8), and the discrete form of (8) is expressed as

$$\begin{aligned} R_x(n, m) &= x\left(n + \frac{2m+a}{2}\right)x^*\left(n - \frac{2m+a}{2}\right) \\ &= \sum_{i=0}^{K-1} A_i^2 \exp(j2\pi f_i(2mT_s + a) \\ &\quad + j2\pi \gamma_i(2mT_s + a)nT_s). \end{aligned} \quad (27)$$

According to [19], the scaling principle is implemented by taking discrete Fourier transform and inverse FFT over n . The scaling transform of $R_x(n, m)$ with Γ operator can be expressed as

$$\begin{aligned} R_x(n', m) &= \frac{1}{N} \sum_{l=-N}^N \sum_{n=-N}^N R_x(n, m) \\ &\quad \cdot \exp\left(-j\frac{2\pi}{N}(2mT_s + a)hnl\right) \exp\left(j\frac{2\pi}{N}n'l\right). \end{aligned} \quad (28)$$

Similarly, the scaling transform of two signals in (12) and (13) with Γ operator can be obtained. Take the discrete Fourier transform to the discrete form of (14) and (15) as

$$\begin{aligned} u(l, m) &= \sum_{n=-N}^N s\left(n + \frac{2m+a}{2}\right) \exp\left(-j\frac{2\pi}{N}(2mT_s + a)hnl\right) \\ &= \sum_{n=-N}^N \sum_{i=0}^{K-1} A_i \exp\left(j2\pi f_i\left(n + \frac{2m+a}{2}\right)T_s\right. \\ &\quad \left.+ j\pi \gamma_i\left(\left(n + \frac{2m+a}{2}\right)T_s\right)^2\right) \\ &\quad \cdot \exp\left(-j\frac{2\pi}{N}(2mT_s + a)hnl\right) \end{aligned} \quad (29)$$

$$\begin{aligned} v(l, m) &= \sum_{n=-N}^N s^*\left(n - \frac{2m+a}{2}\right) \exp\left(-j\frac{2\pi}{N}(2mT_s + a)hnl\right) \\ &= \sum_{n=-N}^N \sum_{i=0}^{K-1} A_i \exp\left(-j2\pi f_i\left(n - \frac{2m+a}{2}\right)T_s\right. \\ &\quad \left.- j\pi \gamma_i\left(\left(n - \frac{2m+a}{2}\right)T_s\right)^2\right) \\ &\quad \times \exp\left(-j\frac{2\pi}{N}(2mT_s + a)hnl\right). \end{aligned} \quad (30)$$

Note that, the peak of the $u(l, m)$ will occur when $l = f_i/(2mT_s + a)h$, and the peak of the $v(l, m)$ will occur when $l = -f_i/(2mT_s + a)h$. To preserve the energy of the signal, we set $-L \leq l \leq L$ where $L \geq f_{\max}/((2mT_s + a)h)$. After

that, take the inverse FFT to (29) and (30) as

$$U(n', m) = \sum_{l=-L}^L u(l, m) \exp\left(j\frac{2\pi}{N}n'l\right) \quad (31)$$

$$V(n', m) = \sum_{l=-L}^L v(l, m) \exp\left(j\frac{2\pi}{N}n'l\right). \quad (32)$$

To guarantee the frequency resolution, (31) and (32) can be rewritten as

$$\begin{cases} U_1(n', 2m-1) = U(n', m) \\ U_1(n', 2m) = U(n', m) \end{cases} \quad (33)$$

$$\begin{cases} V_1(n', 2m-1) = U(n', m) \\ V_1(n', 2m) = U(n', m). \end{cases} \quad (34)$$

Since the centroid frequency window $P_1(\theta)$ is rectangular in this article, the SLVD can be implemented by

$$\begin{aligned} L'(n', 2m') &= \begin{cases} \frac{1}{2\pi} \sum_{m=-f_{th}}^{f_{th}} U_1(n', -m' + m) V_1(n', m' - m), \\ n' = 0 \\ \frac{1}{2\pi} P_2(2m') \sum_{m=-N}^N U_1(n', -m' + m) V_1(n', m' - m), \\ n' \neq 0 \end{cases} \end{aligned} \quad (35)$$

where $P_2(2m') = L'(0, 2m')$. Apply the FFT operator to (35) and the CFCR representation can be obtained.

B. Proposed 3-D Imaging Method

In [13], the 3-D imaging method forms an imaging framework where slant range, Doppler frequency, and chirp rate are estimated together by the FrFT. However, since the pulse repetition frequency and the sampling rate of radar echoes are much lower than those of ladar echoes, a longer imaging time is required to ensure the resolution of the chirp rate in radar imaging. The fast rotation of space debris in a long imaging time yields range migration. In this article, we propose a novel 3-D imaging method for fast-rotating space debris, in which the Khatri–Rao product is operated on the CFCR representation and the RD image to obtain the 3-D image. The flowchart of the proposed 3-D imaging method is shown in Fig. 2.

Firstly, we propose a selection criterion for best imaging time. From (1), slant range of the scattering center is modulated sinusoidal. Since the Doppler frequency and chirp rate is the first and second derivative of the range, they are also modulated sinusoidal. To guarantee that the target's echo is approximated as a multicomponent LFM signal, we choose the best imaging time when the two-order time–frequency moment is the smallest. The two-order time–frequency moment is expressed as (5) in [24]. In the best imaging time, the sum of the Doppler frequency of each scattering center is the smallest, making the echoes approximately linearly modulated. Therefore, the proposed selection criterion for best imaging time

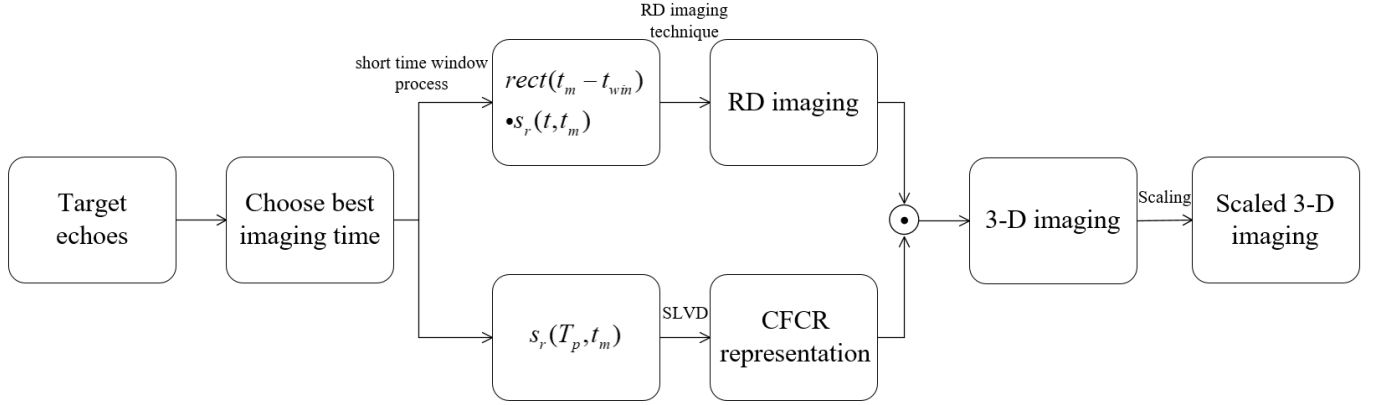


Fig. 2. Flowchart of the proposed 3-D imaging method.

can decrease the influence of range migration by decreasing the gradient of slant range.

Then, we use the RD imaging technique to estimate the slant range. To decrease the influence of range migration, we process a short time window to the RD image. The windowed RD imaging technique can be expressed as

$$S_{RD}(p, q) = \mathcal{F}_{t_m}[\mathcal{F}_t[\text{rect}(t_m - t_{win})s_r(t, t_m)]], \quad S_{RD} \in \mathbb{R}^{P \times Q} \quad (36)$$

where t_{win} denotes the window length, the sampling number in a pulse is P , and the number of pulses in the short time window is Q . The resolution of Doppler frequency relies on the length of imaging time. And the resolution of range depends on the bandwidth of radar instead of the length of imaging time. Therefore, the slant ranges and rough Doppler frequencies of scattering centers can be estimated by finding the peaks of the RD image. The frequency resolution can be calculated by $f_{re} = prf/N_s$ where prf denotes the pulse repletion frequency, and N_s denotes the sampling number. When the frequency resolution is smaller than the frequency interval of different scattering centers, the spectral peaks of different scattering centers are located in the same frequency resolution. In practice, the window length should be artificially adjusted according to the pulse repletion frequency and target motion. Setting the window length too long or too short will result in an error 3-D imaging.

After that, we use the SLVD to obtain the CFCR representation which is expressed as

$$S_{CFCR}(k, q) = \text{SLVD}_{t_m}[s_r(T_p, t_m)], \quad S_{CFCR} \in \mathbb{R}^{K \times Q'} \quad (37)$$

where $\text{SLVD}_{t_m}[\cdot]$ denotes the SLVD operation with respect to t_m , the number of pulses in chosen imaging time is Q' , and $K = 2Q' + 1$. According to the analysis of (4) and (5), when the wavelength is the smallest, the Doppler frequency and chirp rate of scattering centers are the largest. To decrease the relative estimation error, we choose the single frequency signal $s_r(T_p, t_m)$ as $s_t(T_p, t_m) = \exp(j2\pi f_c T_p + j\pi \mu T_p^2)$. From the CFCR representation, we can estimate the centroid Doppler frequencies and chirp rates by searching the peaks. Note that, in the chosen imaging time, Doppler frequency changes fast.

However, we estimate the centroid Doppler frequency which is not influenced by the frequency migration.

Finally, by using the Khatri–Rao product [25], the 3-D image of the target can be realized. Since the Doppler frequency dimension of RD image is different from that of CFCR representation, uniform interpolation along Doppler domain is operated to the RD imaging. The RD imaging after interpolation is expressed as $S'_{RD} \in \mathbb{R}^{P \times Q'}$, in which the row vector is the range domain, and the column vector is the Doppler frequency domain. The CFCR representation is expressed as $S_{CFCR} \in \mathbb{R}^{K \times Q'}$, in which the row vector is the chirp rate domain, and the column vector is the Doppler frequency domain. Note that, both images have the same Doppler frequency domain. In this article, we use the Khatri–Rao product and folding operator to align the RD imaging and the CFCR representation as

$$\begin{aligned} S &= \text{fold}(S'), \quad S \in \mathbb{R}^{P \times Q' \times K} \\ S' &= S'_{RD} \odot S_{CFCR}, \quad S' \in \mathbb{R}^{P \times K \times Q'} \end{aligned} \quad (38)$$

where $\odot$ denotes the Khatri–Rao product operator with respect to the Doppler frequency domain, and $\text{fold}(\cdot)$ denotes the folding operator. The folding operator [26] reshapes $S' \in \mathbb{R}^{P \times K \times Q'}$ into $S \in \mathbb{R}^{P \times Q' \times K}$.

From the peak of RD imaging, we can estimate the range and rough frequency of scattering centers. From the peak of CFCR representation, we can estimate the chirp rate and frequency of scattering centers. Since the peaks of one scattering center in RD imaging and CFCR representation are located in the same frequency resolution unit, we can associate the range and chirp rate estimation by frequency domain. By the Khatri–Rao product along with the frequency domain, the energy of these two peaks is overlapped, and we can estimate the range, frequency, and chirp rate by finding the peak in the obtained 3-D image tensor S . We use the CLEAN technique [27] to filter out this component and estimate the parameters of other scattering centers from the residual echo. Combining with (6), we can obtain the scaled image.

IV. EXPERIMENTS AND PERFORMANCE ANALYSIS

In this section, the performance of the proposed SLVD and the proposed 3-D imaging is evaluated by using the

numerical example and the electromagnetic (EM) simulated data, respectively. In experiment A, we derive the numerical analysis of cross-term and its suppression for the proposed SLVD. In experiment B, the numerical multicomponent LFM signal example is employed to analyze the performance of the proposed SLVD under different signal-to-noise rate (SNR) circumstances. In experiment C, we discuss the choice of centroid frequency window length for SLVD with a numerical example. In experiment D, we choose EM simulated data for two fast-rotating space debris, the triangular pyramid target and the cone-shaped target, to verify the effectiveness of the proposed imaging method.

A. Cross-Term Analysis and Suppression

Here, we give the numerical analysis of the cross-terms in (10). Let $\Delta f_{ij} = (f_i - f_j)/2$, $\Delta \gamma_{ij} = (\gamma_i - \gamma_j)/2$, $\nabla f_{ij} = (f_i + f_j)/2$, and $\nabla \gamma_{ij} = (\gamma_i + \gamma_j)/2$. When $t_n = 0$, the cross-term between i th and j th signal components on the STCF plane can be expressed as

Case 1: $\Delta \gamma_{ij} = 0$

$$\mathcal{F}_\tau(\Gamma[\text{Cr}]) = 2A_i A_j \exp(j2\pi \nabla f_{ij} a) \delta(f - \nabla f_{ij}). \quad (39)$$

Case 2: $\Delta \gamma_{ij} \neq 0$

$$\begin{aligned} \mathcal{F}_\tau(\Gamma[\text{Cr}]) &= A_i A_j \sqrt{\frac{2}{\Delta \gamma_{ij}}} \exp\left(j\frac{\pi}{4}\right) \\ &\times \exp(j4\pi \nabla f_{ij} a) \cos\left(2\pi \frac{(f - \nabla f_{ij})^2}{\Delta \gamma_{ij}}\right). \end{aligned} \quad (40)$$

The proof of (40) and (41) is given in Appendix. We find that the cross-terms on the STCF plane are located at $f_{\text{cr}} = (f_i + f_j)/2$. The proposed SLVD introduces the centroid frequency window to the conventional-form LVD. We consider a multicomponent LFM signal as an example to analyze the suppression of cross-terms

$$\begin{aligned} s_e(t) &= 2 \exp(j2\pi (20t + 5t^2)) \\ &+ 1.5 \exp(j2\pi (-10t - 2.5t^2)). \end{aligned} \quad (41)$$

Through the analysis in (10), the auto-terms are located at $f_{\text{au1}} = 20$ Hz and $f_{\text{au2}} = -10$ Hz. The cross-term is located at $f_{\text{cr}} = (f_{\text{au1}} + f_{\text{au2}})/2 = 5$ Hz.

Fig. 3(a) shows the normalized amplitude in (10) obtained by LVD and (21) obtained by SLVD. The result obtained by LVD has a cross-term at 5 Hz, which can verify the numerical analysis. Fig. 3(b)–(e) shows the normalized amplitude curves of signals $U(0, f)$ and $V(0, F - f)$ in (21) when $F = 0$, f_{au1} , f_{au2} , and f_{cr} . And the window function curve, where the window length is set to $f_{\text{th}} = 5$ Hz, is also shown together. When $F = f_{\text{au1}}$ and $F = f_{\text{au2}}$, the peak amplitudes of these two signals are located in the window. The energy of these two signals is overlapped, and the peak of the auto-terms in Fig. 3(a) can be obtained. When $F = f_{\text{cr}}$, the peaks of these signals are located at the same frequency but not in the window. Therefore, the cross-term is suppressed by the windowing process. Here, we choose $F = 0$ to analyze the sidelobe suppression. In Fig. 3(b), the peak of one signal component is overlapped with the sidelobe of another signal

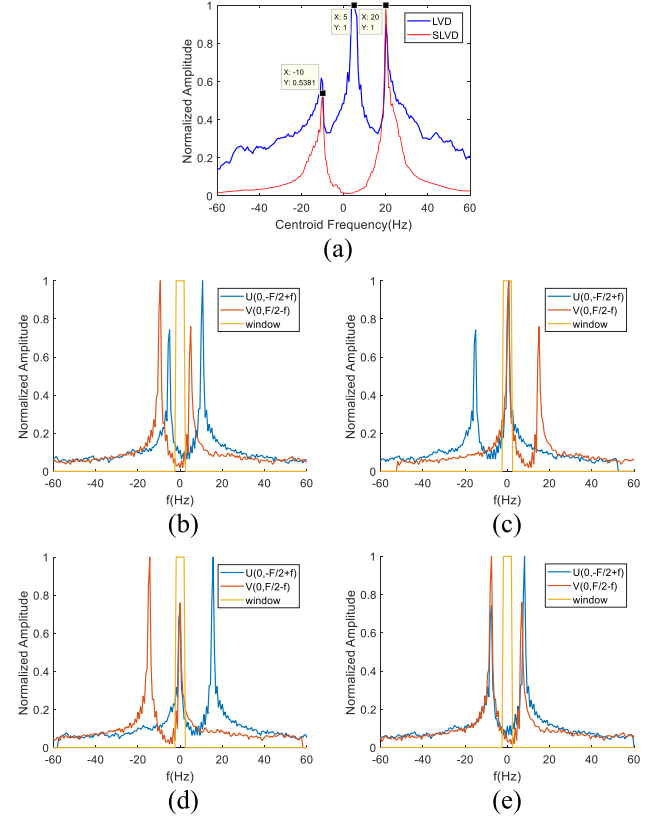


Fig. 3. Cross-term and sidelobe suppression by windowing. (a) Normalized amplitude in (20). (b)–(e) Normalized amplitude for $U(0, -F/2 + f)$, $V(0, F/2 - f)$ when $F = 0$, f_{au1} , f_{au2} , and f_{cr} .

component, which results in the sidelobe in (10). For SLVD, the peaks of the two signal components are not in the window. Therefore, the windowing process can suppress the sidelobes.

B. Performance of SLVD

In this numerical experiment, we operate the LVD and SLVD on (42). The experimental results verify the good performance of the cross-term and sidelobe suppression, especially when SNR is low. For the sake of simplicity, we set $a = 0$ and $h = 1$. The centroid window length $f_{\text{th}} = 5$ Hz, the pulse repetition frequency is 250 Hz, and the dwell time is $T = 0.4$ s. Fig. 4 shows the experimental results when SNR = 20 dB. Fig. 4(a)–(d) shows the CFCR and STCF representations obtained by LVD and SLVD, respectively. And Fig. 4(e) shows the slice of the STCF representation when $t_n = 0$. We can find that the proposed SLVD can effectively suppress cross-terms. Fig. 5 shows the experimental results when SNR = 0 dB to verify the good performance under low SNR circumstance. Fig. 5(a)–(d) shows the CFCR and STCF representations obtained by LVD and SLVD, respectively. And Fig. 5(e) shows the slice of the STCF representation when $t_n = 0$. From Fig. 5(e), we can find that the energy of sidelobes is even higher than that of the auto-terms, and the SLVD can efficiently suppress the sidelobes in STCF representation. The proposed SLVD shows a good performance for suppressing

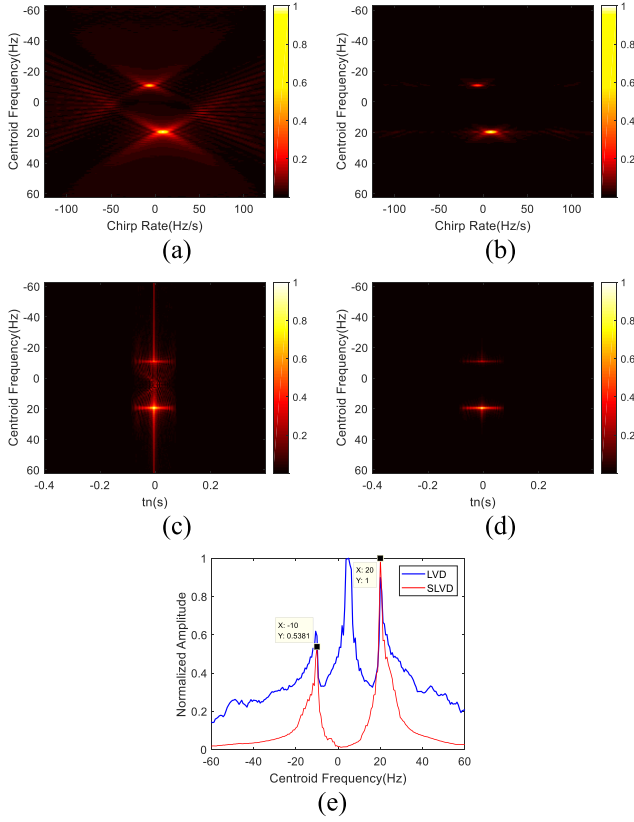


Fig. 4. Experimental results when SNR = 20 dB. (a) CFCR of LVD. (b) CFCR of SLVD. (c) STCF of LVD. (d) STCF of SLVD. (e) Slice of the STCF representation.

cross-terms and sidelobes, especially under low SNR circumstances.

To quantify cross-term suppression performance and noise robustness of the proposed SLVD, we calculate the distribution concentration ratio (DCR) proposed in [19], and we evaluate the estimation accuracy of CFCR by the mean square error (MSE). The formulas are given as follows:

$$\begin{aligned} \text{MSE}_{f_d}(\text{dB}) &= 10 \log_{10}(E[|\hat{f}_d - f_d|^2]) \\ \text{MSE}_{k_d}(\text{dB}) &= 10 \log_{10}(E[|\hat{k}_d - k_d|^2]) \end{aligned} \quad (42)$$

where $\hat{f}_d$ and $\hat{k}_d$ denote estimation of f_d and k_d , respectively. For comparing with the Cramer–Rao lower bounds (CRLBs), we set the units of MSEs to be decibel (dB) instead of meter per second (m/s) and meter per a second squared (m/s²). A higher DCR value indicates a better signal concentration and better cross-term suppression. The DCR of SLVD, LVD, and FRFT is shown in Fig. 6. We can find that DCRs of SLVD are higher than those of LVD and FRFT under different SNRs. Fig. 7 shows the MSEs. The CRLBs can be calculated in [28]. We can find that the MSEs of SLVD are lower than those of LVD and FRFT in low SNR circumstances. The MSEs of estimated frequency and chirp rate obtained by SLVD approach to the CRLBs when SNR is larger than −8 dB. In comparison, the MSEs of estimated frequency and chirp rate obtained by LVD approach to the CRLBs when SNR is −7 dB.

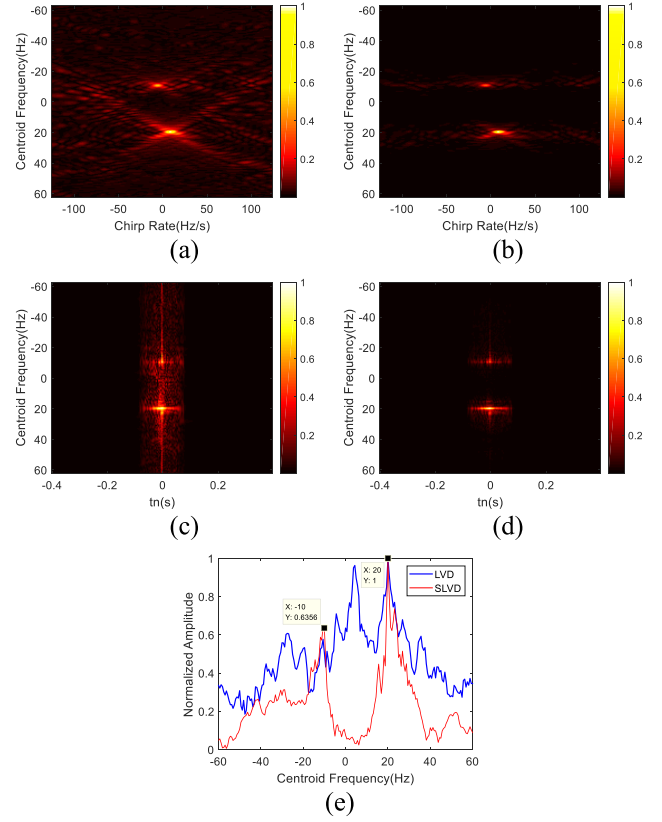


Fig. 5. Experimental results when SNR = 0 dB. (a) CFCR of LVD. (b) CFCR of SLVD. (c) STCF of LVD. (d) STCF of SLVD. (e) Slice of the STCF representation.

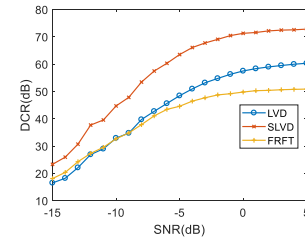


Fig. 6. DCRs by using SLVD, LVD, and FRFT.

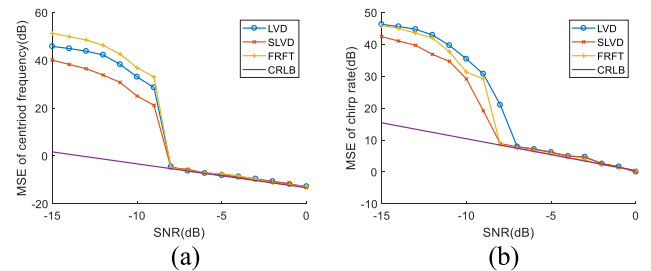


Fig. 7. MSEs by using SLVD, LVD, and FRFT on: (a) centroid frequency and (b) chirp rate.

C. Choice of Centroid Frequency Window Length

In this numerical experiment, we discuss the choice of centroid frequency window length for SLVD. f_{th} denotes half of the window length. When the window length is close

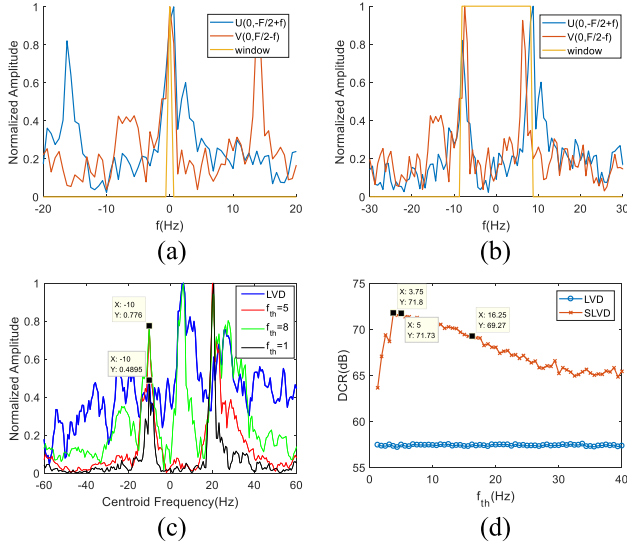


Fig. 8. Analysis for the centroid frequency window length. (a) Normalized amplitude for $U(0, -F/2+f)$, $V(0, F/2-f)$ when $F = Au_2$ and $f_{th} = 1$ Hz. (b) Normalized amplitude for $U(0, -F/2+f)$, $V(0, F/2-f)$ when $F = Cr$ and $f_{th} = 8$ Hz. (c) Normalized slices of STCF under different window lengths. (d) DCRs under different window lengths.

to or larger than the frequency intervals of different signal components, the cross-terms will not be suppressed, but the sidelobes can still be suppressed. When the window length is equal to the frequency scope, SLVD degenerates into LVD. In the low signal-to-noise ratio circumstance, the peaks for $U(0, f)$ and $V(0, f)$ will be broadened or shifted. The energy for auto-terms can be filtered out by the short window. In this experience, the SNR = 0 dB. Fig. 8(a) shows the normalized amplitude for $U(0, -F/2+f)$ and $V(0, F/2-f)$ when $F = Au_2$ and $f_{th} = 1$ Hz. We can find that the peaks are broadened, and part of signal energy is filtered out. Fig. 8(b) shows the normalized amplitude for $U(0, -F/2+f)$, $V(0, F/2-f)$ when $F = Cr$ and $f_{th} = 8$ Hz. It can be concluded that the energy of cross-terms is contained in the window, which makes cross-terms not suppressed. In this article, we set $f_{th} = 5$ Hz. Fig. 8(c) shows normalized slices of STCF for different window lengths. The peak of an auto-term with $f_{th} = 1$ Hz is much lower than that with $f_{th} = 5$ Hz, and the cross-term with $f_{th} = 8$ Hz is not suppressed.

To quantify cross-term suppression performance under different window lengths, we design a 200-times Monte Carlo experiment and calculate the DCRs, which are shown in Fig. 8(d). We can find that the DCR is worse when the window length is too short, the peak DCR gets when $f_{th} = 3.5$ Hz, and the DCR gets worse when the window length grows. Note that, since the sidelobes can still be suppressed, the DCRs are still better than those of LVD.

In practice, we artificially determine the window length by the frequency interval between different signal components. From the peaks for $U(0, f)$ and $V(0, f)$, we can roughly estimate frequencies of signal components. The window length should be a bit shorter than the smallest frequency interval.

Remark: In this remark, we discuss the application scope for the proposed SLVD. SLVD is an LFM parameter estimation

method for mono-component or multicomponent LFM signal. Compared with LVD, the proposed SLVD does not add extra restrictions or parameters except the window length. According to the analysis for the centroid frequency window length, the SLVD has the similar application with these LFM parameter estimation methods [15]–[18], e.g., LVD, FRFT [18], and WVD [15]. For example, in [29] and [30], the product scaled periodic LVD and Radon-LVD are used for coherent integration problem for maneuvering target detection, in which the echoes of maneuvering targets can be modeled as LFM signals. In [31], the modified WVD, which can estimate the coefficients of the polynomial phase, is used for the ISAR imaging with the LFM signal model. The proposed SLVD can also solve these problems introduced above.

D. Performance of the Proposed 3-D Imaging Method

In this subsection, we choose EM simulated data for two fast-rotating space debris to verify the effectiveness of the proposed imaging method. The parameters are set in the following. The carrier frequency is $f_c = 10$ GHz, the bandwidth of radar is 4 GHz, the SNR of echoes is 20 dB, the sampling number in a pulse is $P = 200$, and the pulse repetition frequency is 2000 Hz. The length of imaging time is 0.2 s and the window length of RD imaging is 0.05 s. For simplicity, we set $a = 0$, $h = 1$, and the centroid frequency window length $f_{th} = 15$ Hz.

1) *3-D Imaging for a Triangular Pyramid Target:* In this experiment, we show the 3-D imaging performance of a triangular pyramid target. The geometric model of the target is shown in Fig. 9(a). The 3-D coordinates of scattering centers in the initial time are (0 m, 0 m, 2 m), (1 m, 0 m, 0 m), (0 m, 0.5 m, 0 m), and (−0.5 m, −0.5 m, 0 m). $\theta = 120^\circ$ and $\omega = 4\pi$ rad/s. The RD imaging in a short imaging time is shown in Fig. 9(b), and the CFCR representations obtained by SLVD and LVD are shown in Fig. 9(c) and (d). We can find that the influence of range migration is reduced by chosen imaging time and short time windowing process. However, the frequency resolution is poor in this RD imaging. The CFCR representation can obtain the high resolution estimation of CFCR, and the centroid Doppler frequency which is not influenced by the frequency migration. Comparing Fig. 9(c) and (d), we can find that the proposed SVLD can efficiently suppress the cross-terms and sidelobes. Fig. 9(e) and (f) shows the 3-D imaging before scaling and scaled 3-D imaging, respectively. In comparison, the true positions of scattering centers are also shown in these figures. To quantify the performance of the imaging method, we use estimation errors (ERs) of estimated parameters and relative ERs (RERs) of target size, which can be obtained by

$$\begin{aligned} ER_{f_d} &= |f_d - \hat{f}_d| \\ ER_{k_d} &= |k_d - \hat{k}_d| \\ RER &= |d - \hat{d}|/d \times 100\% \end{aligned} \quad (43)$$

where $\hat{f}_d$ and $\hat{k}_d$ are the estimation result for Doppler frequency and chirp rate. d and $\hat{d}$ denote the true and estimated target size, respectively. In this experiment, we use the edge length to quantify the target size.

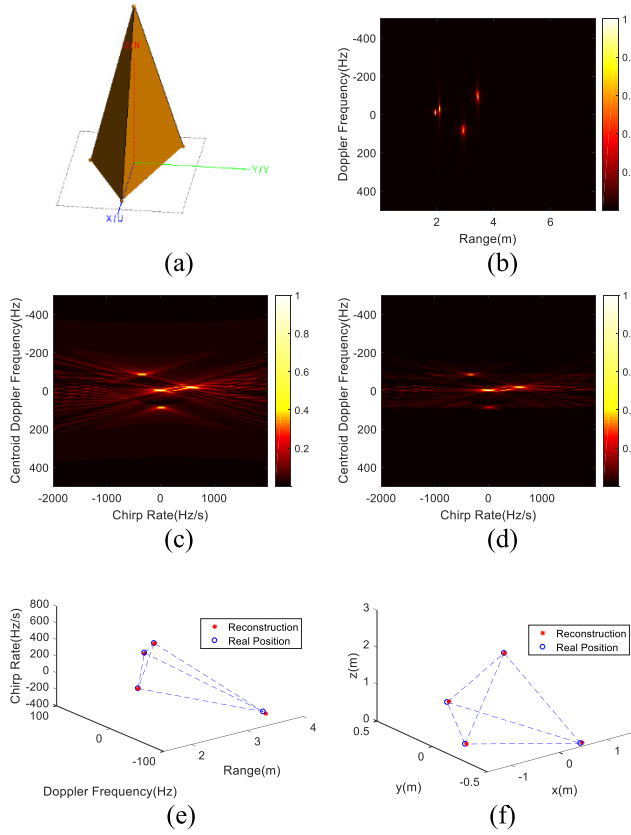


Fig. 9. 3-D imaging performance for triangular pyramid target. (a) Geometric model. (b) RD imaging. (c) CFRCR representation obtained by LVD. (d) CFRCR representation obtained by SLVD. (e) 3-D imaging before scaling. (f) Scaled 3-D imaging.

Table I shows the ERs of Doppler frequency and chirp rate. We compare the estimated results obtained by LVD and SLVD and find that SLVD has a lower chirp rate ER for the first scattering center. Then, we calculate the edge length using the estimated position of scattering centers. Table II shows the RERs of six edge lengths. We can find that the proposed SLVD obtains better estimated results than LVD.

2) 3-D Imaging for Cone-Shaped Target Including Fins:

In this experiment, we show the 3-D imaging performance of the cone-shaped target including fins. Since this target is symmetrical, the Doppler frequencies of different scattering centers are close. The cross-terms and sidelobes may influence the searching for the peak of auto-terms. The geometric model of the target is shown in Fig. 10(a). The 3-D coordinates of scattering centers in the initial time are (0m,0m,2m), (0m,0.75m,-1m), and (0m,0.75m,-1m). $\theta = 120^\circ$ and $w = 4\pi$ rad/s. The RD imaging in a short imaging time is shown in Fig. 10(b), and the CFRCR representations obtained by SLVD and LVD are shown in Fig. 10(c) and (d). We can find that the cross-terms and sidelobes in Fig. 10(c) are higher than those in Fig. 9(c), which will affect the estimated results. Fig. 10(g) shows the slice of the STCF representation to verify the suppression performance more intuitively. We can find that the cross-term, which is located at $f = 16$ Hz, is obviously suppressed. Fig. 10(e) and (f) shows the 3-D imaging before scaling and scaled 3-D imaging, respectively. In comparison,

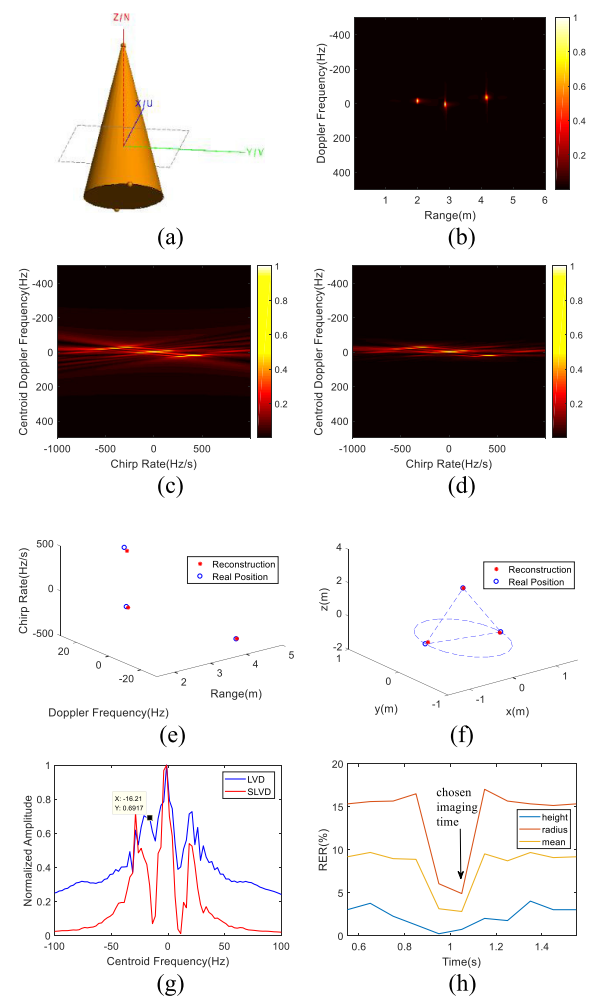


Fig. 10. 3-D imaging performance for cone-shaped target including fins. (a) Geometric model. (b) RD imaging. (c) CFRCR representation obtained by LVD. (d) CFRCR representation obtained by SLVD. (e) 3-D imaging before scaling. (f) Scaled 3-D imaging. (g) Slice of the STCF representation when $t_n = 0$. (h) Imaging performance in different imaging times.

the true positions of scattering centers are also shown in these figures. In this experiment, we use the height and radius of the bottom plane to quantify the target size.

Table III shows the ERs of Doppler frequency and chirp rate. Table IV shows the RERs of high and radius. For LVD, we can find that a serious error happens for estimating the first scattering center's chirp rate, which makes the RERs of high and radius obtained by LVD far worse than those obtained by SLVD. The proposed method shows a good performance on estimation, where the RERs are lower than 5%.

In this experiment, the window length is still a bit shorter than the Doppler frequency interval, which induces effective suppression. However, the proposed algorithm will be not ideal when the Doppler frequencies of scattering centers are too dense. The window length should be shorter than the frequency interval between different scattering centers. When the Doppler frequencies of scattering center are dense, the window length should be too short, which filters the auto-terms' energy. Therefore, it is difficult to make a compromise between retaining the auto-terms' energy and suppression for

TABLE I
ERS OF ESTIMATED PARAMETERS FOR A TRIANGULAR PYRAMID TARGET

scattering center			1	2	3	4	Mean
Doppler frequency	LVD	real	-81.7808	-14.2773	0	88.9194	-
		estimated	-83.5411	-16.2095	-1.2469	88.5287	-
		ER	1.7604	1.9321	1.2469	0.3907	1.3325
	SLVD	estimated	-83.5411	-16.2095	-1.2469	88.5287	-
		ER	1.7604	1.9321	1.2469	0.3907	1.3325
		real	-301.800	558.743	0	22.4286	-
chirp rate	LVD	estimated	-334.164	563.591	4.9875	14.9626	-
		ER	32.3643	4.8476	4.9875	7.4660	12.4164
		estimated	-324.189	563.591	4.9875	14.9626	-
	SLVD	ER	22.3892	4.8476	4.9875	7.4660	9.9226
		real	-301.800	558.743	0	22.4286	-
		estimated	-334.164	563.591	4.9875	14.9626	-

TABLE II
RERS OF EDGE LENGTHS FOR A TRIANGULAR PYRAMID TARGET

edge		1	2	3	4	5	6	Mean
LVD	real	1.5811	2.1213	1.1180	2.2361	1.1180	2.0616	-
	estimated	1.6189	2.1301	1.1294	2.2619	1.1228	2.0710	-
	RER	2.39%	0.41%	1.01%	1.16%	0.43%	0.46%	0.98%
	SLVD	estimated	1.6016	2.1301	1.1201	2.2270	1.1228	2.0413
SLVD	RER	1.29%	0.41%	0.18%	0.40%	0.43%	0.98%	0.62%

TABLE III
ERS OF ESTIMATED PARAMETERS FOR A CONE-SHAPED TARGET INCLUDING FINS

scattering center			1	2	3	Mean
Doppler frequency	LVD	real	-20.7246	0	20.7246	-
		estimated	-28.4683	-1.2469	18.7032	-
		ER	7.9537	1.2469	2.0214	3.7406
	SLVD	estimated	-21.1970	-1.2469	18.7032	-
		ER	0.4724	1.2469	2.0214	1.2469
chirp rate	LVD	real	-415.332	0	415.332	-
		estimated	-306.733	2.4938	396.508	-
		ER	108.599	2.4938	18.8240	43.3058
	SLVD	estimated	-406.483	2.4938	396.508	-
		ER	8.8489	2.4938	18.8240	10.055

TABLE IV
RERS FOR A CONE-SHAPED TARGET INCLUDING FINS

size		height	radius	Mean
	real	3	0.75	-
LVD	estimated	3.1289	1.2615	-
	RER	4.3%	15.9%	10.1%
SLVD	estimated	2.9773	0.7131	-
	RER	0.76%	4.92%	2.84%

cross-terms. Note that, the smoothed window is only operated on the frequency domain. Dense slant ranges or dense chirp rates will not influence the performance of the proposed SLVD.

To quantify the imaging performance during different imaging times, we calculate the RERs for target height and radius in Fig. 10(h). We can find that the mean RER in the chosen imaging time is better than those in other time, which verify the effectiveness of the proposed selection criterion for best imaging time.

V. CONCLUSION

In this article, we propose a novel 3-D imaging method based on SLVD. Firstly, the selection criterion for best imaging time based on time–frequency moment is proposed. Then, we obtain the 3-D image by operating the Khatri–Rao product on the CFCR representation and the RD imaging. We process a short time window during the RD imaging procedure to decrease the influence of range migration. And the SLVD is proposed to obtain the CFCR representation for cross-term and sidelobe suppression. We give the analytical formula combined with a numerical experiment to demonstrate the effectiveness for cross-term and sidelobe suppression. In addition, EM experimental results demonstrate that the proposed imaging method is capable of obtaining 3-D images of fast-rotating space debris and acquiring more accurate estimation of target size. In future work, the proposed SLVD is an LFM estimation method, which can be commonly used in other problems, such as coherent integration problems. The proposed imaging method is suitable for space debris, which has a few scattering centers. The 3-D imaging for complicated space targets, such as satellites, is the future work.

APPENDIX

In the scaled PSIAF in (9), the cross-term between the i th and j th signal component can be expressed as

$$\Gamma[\text{Cr}] = A_i A_j R^{C_1} \Re(2R^{C_2}) \quad (44)$$

with

$$R^{C_1} = \exp(j2\pi \nabla f_{ij} \tau' + j2\pi \nabla \gamma_{ij} t \tau') \quad (45)$$

$$R^{C_2} = \exp\left(j2\pi \Delta f_{ij} \frac{t_n}{h\tau'} + j2\pi \Delta \gamma_{ij} \left(\frac{t_n}{h\tau'}\right)^2 + j2\pi \Delta \gamma_{ij} \frac{\tau'^2}{4}\right) \quad (46)$$

where $\Re$ denotes the operation of taking real part and $\tau' = t + a$.

Case 1: $\Delta \gamma_{ij} = 0$. Equation (44) can be simplified as

$$\Gamma[\text{Cr}] = 2A_i A_j \exp\left(j2\pi \nabla f_{ij} \tau' + j2\pi \nabla \gamma_{ij} \frac{t_n}{h}\right) \times \cos\left(4\pi \Delta f_{ij} \frac{t_n}{h\tau'}\right). \quad (47)$$

When $t_n = 0$, the cross-term on the STCF plane can be obtained by

$$\mathcal{F}_\tau(\Gamma[\text{Cr}]) = 2A_i A_j \exp(j2\pi \nabla f_{ij} a) \delta(f - \nabla f_{ij}). \quad (48)$$

Case 2: $\Delta \gamma_{ij} \neq 0$. When $t_n = 0$, (44) can be rewritten as

$$\Gamma[\text{Cr}] = 2A_i A_j \exp(j2\pi \nabla f_{ij} \tau') \cos\left(2\pi \Delta \gamma_{ij} \frac{\tau'^2}{4}\right). \quad (49)$$

The Fourier transform is performed on the first term in (49) and we can obtain

$$\mathcal{F}_\tau(\exp(j2\pi \nabla f_{ij} \tau')) = \exp(j2\pi \nabla f_{ij} a) \delta(f - \nabla f_{ij}). \quad (50)$$

The second term in (49) is expressed as

$$\begin{aligned} & \cos\left(2\pi \Delta \gamma_{ij} \frac{\tau'^2}{4}\right) \\ &= \frac{1}{2} \left[\exp\left(j2\pi \Delta \gamma_{ij} \frac{\tau'^2}{4}\right) + \exp\left(-j2\pi \Delta \gamma_{ij} \frac{\tau'^2}{4}\right) \right]. \end{aligned} \quad (51)$$

The Fourier transform is performed on the first term in (51) and we can obtain

$$\begin{aligned} & \mathcal{F}_\tau\left(\exp\left(j2\pi \Delta \gamma_{ij} \frac{\tau'^2}{4}\right)\right) \\ &= \int_{-\infty}^{\infty} \exp\left(j2\pi \Delta \gamma_{ij} \frac{\tau'^2}{4}\right) \exp(-j2\pi f \tau) d\tau. \end{aligned} \quad (52)$$

Similar to the cross-term analysis in [19], the principle of stationary phase in [32, Ch. 2] is used. The stationary point τ^* is determined by

$$\frac{du_1(\tau)}{d\tau} = \frac{d}{d\tau} \left(\text{phase}\left(\exp\left(j2\pi \Delta \gamma_{ij} \frac{\tau'^2}{4}\right)\right) - 2\pi f \tau \right) = 0. \quad (53)$$

We have a stationary point $\tau^* = (2f/\Delta \gamma_{ij}) - a$ and its derivatives evaluated at this stationary point $(d^2 u_1(\tau)/d\tau^2)|_{\tau=\tau^*} = (\pi \Delta \gamma_{ij}/f)$.

Then, we can calculate (52) by

$$\begin{aligned} & \mathcal{F}_\tau\left(\exp\left(j2\pi \Delta \gamma_{ij} \frac{\tau'^2}{4}\right)\right) \\ &= \sqrt{\frac{2}{\Delta \gamma_{ij}}} \exp\left(j\frac{\pi}{4}\right) \exp\left(j2\pi f \left(-\frac{f}{\Delta \gamma_{ij}} + a\right)\right). \end{aligned} \quad (54)$$

In the same way, we can obtain

$$\begin{aligned} & \mathcal{F}_\tau\left(\exp\left(-j2\pi \Delta \gamma_{ij} \frac{\tau'^2}{4}\right)\right) \\ &= \sqrt{\frac{2}{\Delta \gamma_{ij}}} \exp\left(j\frac{\pi}{4}\right) \exp\left(j2\pi f \left(\frac{f}{\Delta \gamma_{ij}} + a\right)\right). \end{aligned} \quad (55)$$

According to (50), (54), and (55), the operation of Fourier transform on (49) can be expressed as

$$\begin{aligned} \mathcal{F}_\tau[\text{Cr}] &= A_i A_j \mathcal{F}_\tau[\exp(j2\pi \nabla f_{ij} \tau')] \otimes_\tau \\ & \left[\mathcal{F}_\tau\left(\exp\left(j2\pi \Delta \gamma_{ij} \frac{\tau'^2}{4}\right)\right) \right. \\ & \quad \left. + \mathcal{F}_\tau\left(\exp\left(-j2\pi \Delta \gamma_{ij} \frac{\tau'^2}{4}\right)\right) \right] \\ &= A_i A_j \sqrt{\frac{2}{\Delta \gamma_{ij}}} \exp\left(j\frac{\pi}{4}\right) \exp(j4\pi \nabla f_{ij} a) \\ & \quad \times \cos\left(2\pi \frac{(f - \nabla f_{ij})^2}{\Delta \gamma_{ij}}\right). \end{aligned} \quad (56)$$

Here, the proof of the cross-terms on the STCF plane is completed by (48) and (56).

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